Smart Meeting Assistant: A Shared-Transcript Modular Pipeline with an LLM-as-Judge Pilot for Bilingual Chinese–English Meetings

CS4300 NLP Group Project (Topic 5)
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Abstract

We present an end-to-end smart meeting assistant for bilingual (Chinese–English) settings that unifies five participant-facing modules—ASR with speaker diarization, summarization, translation, action-item extraction, and sentiment / engagement analysis—over a single shared-transcript representation, with a retrieval-augmented Q&A interface and a live-microphone mode. Each language-understanding module pairs a GPT-4o-mini instantiation with a task-specific pretrained baseline (BART-large-CNN, NLLB-200 distilled 600 M), and we run a small LLM-as-Judge pilot to complement ROUGE/BLEU with Likert ratings of faithfulness, coverage, and specificity. On AMI ES2004a, AliMeeting R8001_M8004, MELD, and 26 news-commentary EN↔ZH pairs, the LLM matches or exceeds the specialized baselines on surface metrics; the judge pilot surfaces a faithfulness-vs-coverage trade-off between the long-context LLM and the 1 024-token BART that ROUGE alone hides. We report honest per-direction translation numbers and a task-detection-only action-item evaluation, and discuss the system's current limits.

1 Introduction

Meetings in Hong Kong academic, legal, and business settings routinely switch between Cantonese, Mandarin, and English. Participants face a compound cognitive load: following the spoken content, mentally translating across languages, tracking action items, and sensing the room's emotional dynamic. Commercial tools (Zoom AI Companion, Otter.ai) focus on English and provide post-hoc summaries rather than in-session support, and most open-source meeting NLP research has concentrated on single-language benchmarks such as AMI [1].

We build a smart meeting assistant that explicitly addresses the bilingual setting and provides both in-session streaming support and post-hoc structured analysis. Given a meeting audio input the system produces six participant-facing views: a speaker-labeled transcript, an abstractive summary with decisions and follow-ups, a translated transcript in the complementary language, a table of action items grounded in transcript spans, a sentiment-and-engagement timeline, and an embedding-retrieval Q&A interface. A seventh live mode streams microphone input through chunked Whisper transcription and asynchronous GPT-4o-mini translation.

Our contributions are:

(i) A **shared-transcript modular architecture** in which WhisperX with pyannote diarization emits a canonical transcript JSON that four downstream LLM modules consume independently, enabling independent evaluation and ablation.

(ii) **Bilingual-aware prompting**: WhisperX emits one meeting-level language tag, and a CJK character-ratio heuristic is used as a per-utterance fallback only when that tag is `auto`; on top of that we apply a speaker-constrained function-calling schema for action-item extraction that forces the LLM to choose assignees from the upstream speaker set.

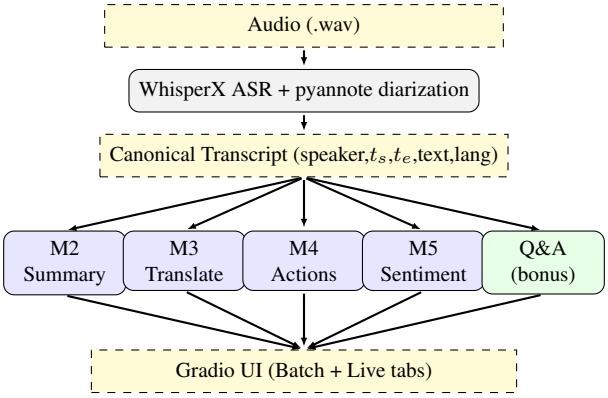


Figure 1: Shared-transcript modular architecture. A single WhisperX pass produces the canonical transcript JSON that every downstream module consumes. Yellow = data artifacts, grey = pretrained models, blue = GPT-4o-mini calls, green = retrieval-based bonus.

(iii) An **LLM-as-Judge pilot** (three triples, summarization and actions) that complements ROUGE/BLEU with Likert ratings on faithfulness, coverage, and specificity.

(iv) A **live streaming mode** that combines `faster-whisper base` chunked ASR with a thread-pool of asynchronous translation LLM calls on commodity laptops.

2 Related Work

Speech recognition and diarization. Whisper [2] established strong zero-shot ASR via 680 k h of weakly supervised training. pyannote.audio [3] with the powerset formulation [4] remains the standard open-source diarization stack; WhisperX combines the two with wav2vec2 forced alignment. We use these as unmodified building blocks.

Meeting corpora and baselines. AMI [1] and AliMeeting (M2MeT) [5] are the canonical English and Mandarin multi-party meeting corpora; MELD [6] supplies 7-way conversational emotion labels. On the generation side, BART [9] is the default encoder-decoder summarizer (1 024-token limit, decisive on meeting-length inputs), and NLLB-200 [7] is the de facto MT baseline—its distilled 600 M checkpoint performs markedly worse on EN→ZH than on ZH→EN in our benchmark (Sec. 4).

LLM-based evaluation. Zheng *et al.* [8] score Likert ratings from GPT-4 to approximate human judgements, reporting $\sim$80% agreement with crowdworkers. We adopt their framework on faithfulness / coverage / specificity and use GPT-4o (not our system's GPT-4o-mini) as the judge to limit self-preference bias.

3 Methodology

3.1 Shared-Transcript Modular Architecture

Fig. 1 shows the pipeline. A single WhisperX pass emits a canonical `Transcript` JSON object—a list of `Utterance` records carrying `(speaker,`t_s`,`t_e`,text,lang)`—that all four downstream modules and the Q&A index consume without re-running ASR. This factoring has three payoffs: (i) the ASR and diarization cost of roughly $0.5\times$ real-time on CPU is paid once per audio file (then cached by SHA-256); (ii) the downstream modules can be evaluated and ablated independently; and (iii) introducing a new module (here, Q&A and live mode) requires no changes to upstream components. The four language-understanding modules are independent in the DAG sense but are currently executed serially; the architecture does not prevent a parallel driver.

3.2 Bilingual-Aware Prompting

The translation module (M3) performs per-utterance ZH↔EN translation with a three-turn sliding context window of prior utterances, enabling pronoun and antecedent resolution across speaker turns. Language handling is *meeting-level* by default: WhisperX stamps one language tag on every utterance, and our CJK character-ratio heuristic only fires as a fallback when that tag is `auto`. This is adequate for the monolingual AMI and AliMeeting meetings we evaluate, but a proper per-utterance LID (e.g., VoxLingua) would be needed for real code-switched Hong Kong meetings; we leave that to future work. The summary module (M2) detects the dominant language from the transcript and emits bullets in that language, so a Mandarin meeting yields a Mandarin summary without extra configuration.

3.3 Structured Action-Item Extraction

M4 uses OpenAI function calling with a strict JSON schema. The `assignee` field is declared as an `enum` populated with the speaker IDs produced by M1, so the model cannot invent a non-existent participant as owner. The schema further requires an `evidence_span`—a direct quote from the transcript—so every extracted action is verifiable against its source line:

```
{ "items": [{
    "task": str, "assignee": one_of(speakers),
    "due": str|null, "priority": "H"|"M"|"L",
    "evidence_span": str }] }
```

3.4 Live Streaming Mode

The live tab captures microphone audio via Gradio's streaming callback (500 ms delta chunks). Our `MicStream` class accumulates a 1-second buffer, runs `faster-whisper base` (int8, CPU) on the buffer, and dispatches each resulting utterance to a thread-pool worker for the translation LLM call. Live mode does *not* run pyannote diarization—it is too heavy for 1-s chunks on CPU—so utterances are emitted under a single `SPEAKER` label; the batch pipeline retains full speaker attribution. A **Process now** button force-flushes residual audio when the speaker pauses below the buffer threshold.

3.5 LLM-as-Judge Pilot

ROUGE and BLEU are n-gram surface-form metrics, and BART's 1 024-token truncation is known to depress them on meeting-length inputs. We therefore run a small LLM-as-Judge pilot (3 triples: GPT summary, BART summary, and GPT action-item list—all on the AMI ES2004a full meeting) where GPT-4o rates each {transcript, output, system} triple on three 1–5 Likert axes: *faithfulness* (does the output avoid hallucination?), *coverage* (is important content captured?), and *specificity* (is the output concrete rather than vague?). We cap the transcript shown to the judge at 8 000 characters to fit its context budget. This is a pilot, not a full human-correlation study, and its sample size is explicitly small; Sec. 5 explains how the signal still complements surface metrics.

4 Experiments

4.1 Datasets

Table 1 lists the evaluation data. All are public; no component is trained on these sets.

4.2 Module 1 — ASR with Diarization

We report Word Error Rate for English (AMI ES2004a) and Character Error Rate for Chinese (AliMeeting 2-min clip), both with a lowercase-and-punctuation-stripped normalization applied identically to hypothesis and reference (Table 2). For diarization we use pyannote-3.1 unmodified; the underlying

Table 1: Evaluation datasets. Sizes are the portions actually scored in this paper.

Dataset	Lang.	Use	Size used
AMI ES2004a [1]	EN	M1–M4	30-min meeting
AliMeeting R8001_M8004 [5]	ZH	M1 CER	2-min clip
MELD dev [6]	EN	M5	150 utts
news_commentary	EN↔ZH	M3 BLEU	13×2 pairs

diarization formulation is described in [4]. We do *not* report Diarization Error Rate in this paper—our pipeline change is limited to the integration of that third-party model into WhisperX, and scoring DER against AMI's RTTM labels is left to future work. The elevated WER reflects Whisper's discourse normalization (Sec. 5).

Table 2: ASR baseline numbers (WhisperX large-v3, int8 CPU).

Dataset	Metric	Value
AMI ES2004a (30 min)	WER	0.407
AliMeeting R8001_M8004 (2-min clip)	CER	0.325

4.3 Module 2 — Summarization

We compare GPT-4o-mini against BART-large-CNN on two regimes: the first two minutes of ES2004a (short-clip ablation) and the full 30-minute meeting. ROUGE is computed against the AMI reference abstract (Fig. 2). GPT's 128 K-token context handles the full meeting; BART's 1 024-token limit forces truncation, so its full-meeting ROUGE-1 actually drops below its own short-clip ceiling while GPT's rises. The ROUGE-1 gap widens from $+0.066$ on the clip to $+0.230$ on the full meeting.

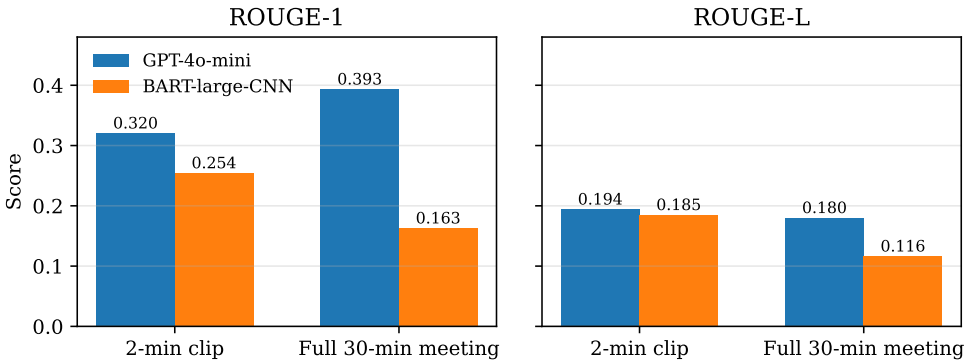


Figure 2: Summary ROUGE against AMI abstract, 2-min clip vs full 30-min meeting. BART's 1 024-token context limit is decisive.

4.4 Module 3 — Translation

We sample 13 EN–ZH pairs from `Helsinki-NLP/news_commentary` and evaluate in both directions (26 hypotheses total), each direction scored with its sacreBLEU-appropriate tokenizer (`zh` for ZH targets, `13a` for EN targets) following [10]. Our batch pipeline translates each utterance with a three-turn context window; the BLEU benchmark translates isolated sentences (i.e., no context) and is therefore a conservative proxy for the system's full in-meeting quality. Fig. 3 reports the per-direction scores.

4.5 Module 4 — Action-Item Detection

AMI's abstractive summary annotations include a machine-readable `<actions>` section. We score our function-calling output against those gold actions using greedy cosine-similarity matching over sentence-transformer embeddings (threshold 0.55). ES2004a annotates one gold action and no gold

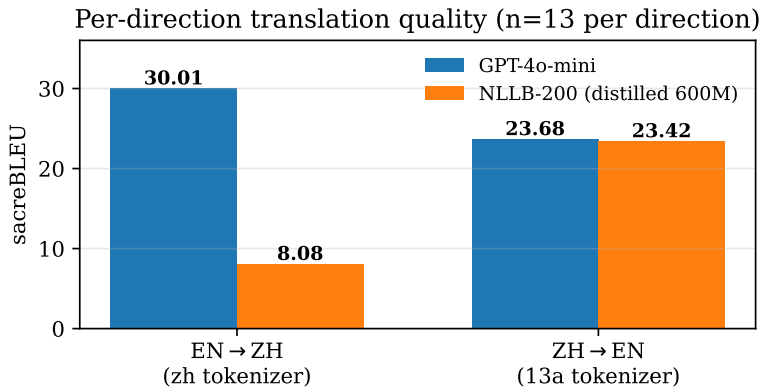


Figure 3: Per-direction sacreBLEU on news-commentary sentences (isolated-sentence setting). GPT-4o-mini dominates NLLB-200 (distilled 600 M) on EN→ZH; the two systems are numerically close on ZH→EN (23.68 vs 23.42 BLEU on 13 pairs).

assignees; our system extracts four items, the one gold task plus three procedural routine items (Table 3). This evaluation therefore measures *task detection only*. Structured fields (`due`, `priority`, and `evidence_span`) are produced and human-inspectable in the UI but are not scored against gold here, because AMI does not annotate them.

Table 3: Action-item task-detection on AMI ES2004a (1 gold task).

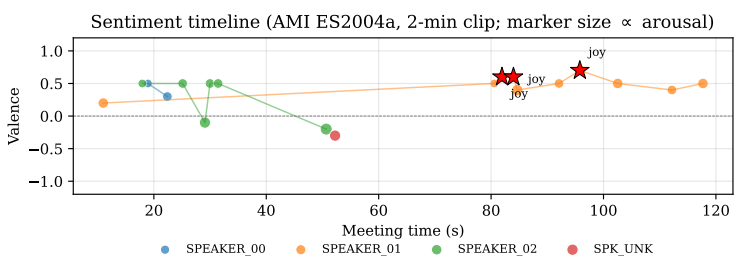
Precision	Recall	F1
0.25	1.00	0.40

4.6 Module 5 — Sentiment / Emotion

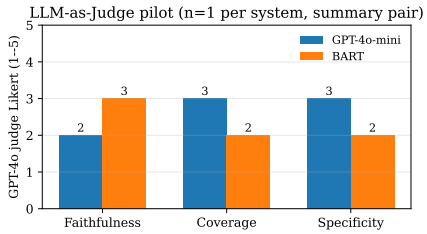
Per-utterance 7-way emotion classification on 150 randomly sampled MELD *dev* utterances yields **0.580 accuracy / 0.519 macro-F1**. This is a zero-shot result with no task-specific fine-tuning and no dialogue context. Supervised context-aware systems on MELD typically reach 0.65–0.70 accuracy; our number is competitive for the zero-shot, single-utterance setting we evaluate, and we do not yet test on Chinese meetings—a clear next step. Fig. 4(a) visualises the valence timeline produced by the module on ES2004a.

4.7 LLM-as-Judge

GPT-4o rates the three pilot triples from Sec. 3.5 (GPT summary, BART summary, GPT actions) on faithfulness / coverage / specificity. Fig. 4(b) visualises the summarization pair.



(a) Sentiment timeline (ES2004a, 2-min clip).



(b) GPT-4o judge Likert (pilot, n=1 per system).

Figure 4: (a) Per-utterance sentiment from `pipeline_run.json`; marker size encodes arousal, red stars mark the top-3 moments ranked by |valence| + arousal (the same formula used in `src/sentiment.py`). (b) Judge pilot: BART is scored more *faithful* on the truncated input it sees; GPT scores higher on *coverage* and *specificity*—a trade-off surface ROUGE hides.

5 Discussion

BART context bottleneck dominates long meetings. The ROUGE-1 gap between GPT-4o-mini and BART widens from $+0.066$ on a 2-minute clip to $+0.230$ on the full 30-minute meeting (Fig. 2). BART's 1 024-token truncation forces it to see only the first few minutes of the meeting, so our short-clip numbers effectively cap its full-meeting ceiling. Long-context LLMs are not just convenient here—they are structurally necessary.

Direction asymmetry in translation. The per-direction BLEU reveals a story a single corpus-BLEU would flatten. On our 13 EN→ZH pairs, GPT-4o-mini scores 30.01 against NLLB's 8.08; on the 13 ZH→EN pairs the two systems score 23.68 and 23.42 respectively, a 0.26 BLEU gap that we do *not* interpret as meaningful on so few pairs and without resampling-based intervals. The EN→ZH asymmetry is consistent with the known weakness of distilled NLLB on generate-Chinese directions. BLEU does not aggregate linearly across target languages, so we report the two directions side-by-side and avoid a single summary score.

Surface metrics miss faithfulness. Both ROUGE-1 and ROUGE-L nominally favour GPT, yet the judge pilot (Fig. 4(b)) finds BART *more faithful* on the truncated input it sees; GPT's advantage is in covering more of the meeting and producing more concrete phrasing. A summary system optimized purely for ROUGE would not surface this behaviour. The pilot is small (three triples) and we therefore frame it as a qualitative diagnostic rather than a claim of statistical superiority.

WER as a misleading quality proxy. Our 0.407 WER on AMI is markedly worse than Whisper's reported $\sim$0.15 on clean read speech, but much of the gap is discourse normalization ("she gonna answer me" in the AMI literal transcript becomes "Is she going to answer me?" in Whisper output). For downstream understanding (M2–M5) this is arguably an improvement; normalization-aware WER would be more informative.

Live-mode latency budget. Manual stopwatch observation on a single Apple-Silicon MacBook places end-to-end latency from mouth-closed to on-screen translation at roughly 2–4 s, dominated by the 1-second audio buffer and a 1–2 s `faster-whisper base` CPU pass (translation is asynchronous). This is an eyeballed estimate, not an instrumented measurement; it is sufficient for "near-real-time" participant support but not truly simultaneous interpretation, which would need GPU-streaming ASR or a server-side Whisper API. Adding a latency logger is on our short-term to-do list.

Scope and small-N caveats. Most numbers come from a single AMI meeting, and the judge pilot scores only three triples. The data-prep scripts in `eval/data_prep/*.py` already scale to dozens of meetings, and Chinese-meeting evaluation beyond a 2-minute CER spot check is left to future work. We do not score DER: pyannote-3.1 is used unmodified [4] and scoring it against AMI's RTTM labels is left to future work.

6 Conclusion

We built a bilingual smart meeting assistant organised around a shared-transcript modular architecture, with four GPT-4o-mini modules, retrieval-augmented Q&A, and a live streaming mode. GPT-4o-mini matches or exceeds BART and NLLB-200 on meeting summarization and EN→ZH translation, and the LLM-as-Judge pilot surfaces a faithfulness / coverage trade-off that surface metrics hide. Future work: (i) scale to the full AMI/AliMeeting partitions and score DER; (ii) swap the live ASR backend to GPU-streaming or a server Whisper API; (iii) replace the CJK-ratio heuristic with a proper per-utterance LID for code-switched meetings; and (iv) expand the judge study to include human correlation.

References

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A Prompts used

Full system prompts for every module are shipped in the source tree:

- Summarization: `src/summarize.py`
- Translation: `src/translate.py`
- Action items: `src/actions.py`
- Sentiment: `src/sentiment.py`
- Q&A: `src/qa.py`
- LLM judge: `eval/llm_judge.py`

B Presentation slides

The 15-minute presentation slides will be appended here in the final submission PDF, as required by the project brief.